

QUANTITATIVE RESEARCH REPORT

# AI-Powered Trading & Risk Analytics System

*A comprehensive end-to-end quantitative finance system covering algorithmic strategy design, machine learning, portfolio optimisation, and institutional-grade risk analytics.*

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Portfolio  
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Risk Analytics

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# 1 | EXECUTIVE SUMMARY

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This report documents the design, development, and validation of an **AI-Powered Trading and Risk Analytics System** — a full-stack quantitative finance platform built entirely in Python. The system operationalises the core workflow of a modern proprietary trading desk: ingesting real-time market data, engineering predictive features, generating trade signals through rule-based strategies and a machine learning layer, rigorously backtesting those signals, optimising capital allocation across a multi-asset portfolio, and quantifying tail risk using institutional-grade metrics.

The platform was applied to a universe of five large-cap U.S. equities (AAPL, MSFT, GOOGL, AMZN, TSLA) over a five-year lookback period (January 2020 – December 2024). In simulation, the Moving Average Crossover strategy delivered a Sharpe ratio of 1.84 and a maximum drawdown of  $-14.2\%$ , comfortably outperforming the buy-and-hold benchmark on a risk-adjusted basis. The Markowitz-optimised portfolio achieved an annualised return of  $19.7\%$  with a portfolio volatility of  $13.6\%$ , yielding an efficient frontier that reveals meaningful diversification gains beyond equal-weighting.

The system is delivered as a modular Python codebase accompanied by an interactive Streamlit dashboard, enabling both technical and non-technical stakeholders to explore strategy performance, portfolio composition, and risk exposure in real time. The architecture is designed for extensibility: each module can be independently upgraded — for example, replacing the Linear Regression baseline with an LSTM or Transformer model — without refactoring the surrounding pipeline.

**Key finding: On a risk-adjusted basis, the AI-augmented MA Crossover strategy outperformed passive buy-and-hold by 6.3 percentage points of annualised Sharpe-equivalent return over the 2020–2024 backtest window, with 22% lower maximum drawdown.**

# 2 | PROBLEM STATEMENT

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Retail and institutional investors face a common set of structural challenges when deploying capital in equity markets. First, **information asymmetry**: sophisticated market participants with access to proprietary data, co-located servers, and quantitative talent systematically capture alpha that is invisible to those relying on manual analysis. Second, **behavioural bias**: discretionary traders are susceptible to loss aversion, anchoring, and recency bias — distortions that rules-based algorithmic systems are specifically designed to eliminate. Third, **risk opacity**: most retail portfolios are constructed without formal risk metrics, meaning that managers frequently discover their true drawdown exposure only after a loss event has occurred.

Meanwhile, the barriers to building a serious quantitative system have fallen dramatically. Open-source libraries (pandas, scikit-learn, TensorFlow), free market data APIs (Yahoo Finance, Alpha Vantage), and cloud computing have democratised the tooling previously available only to bulge-bracket banks and well-capitalised hedge funds. Yet most publicly available trading tutorials remain superficial — demonstrating toy strategies on clean data without addressing the realities of transaction costs, look-ahead bias, regime changes, or tail risk.

This project bridges that gap. It constructs a system that is simultaneously **rigorous enough for professional scrutiny** and **accessible enough for a single developer** to maintain and extend. The central

question it answers is: *Can a systematic, data-driven approach — combining classical technical analysis, machine learning, and modern portfolio theory — generate superior risk-adjusted returns compared to passive equity exposure?*

### 3 | OBJECTIVES

| Objective                           | Description                                                                                                                                                                |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>O-1: Data Infrastructure</b>     | Build a robust, cacheable data ingestion pipeline capable of fetching, cleaning, and storing OHLCV data for any liquid equity or ETF via the Yahoo Finance API.            |
| <b>O-2: Feature Engineering</b>     | Derive a rich feature set including trend indicators (SMA), momentum oscillators (RSI), volatility proxies (ATR), and lagged return sequences suitable for ML consumption. |
| <b>O-3: Strategy Implementation</b> | Implement two production-style trading strategies — Moving Average Crossover and RSI Mean Reversion — with clearly defined entry, exit, and position-sizing rules.         |
| <b>O-4: ML Augmentation</b>         | Train a Linear Regression baseline and an LSTM sequence model to predict next-day price direction, integrating their signals into the strategy layer.                      |
| <b>O-5: Backtesting</b>             | Design a vectorised, bias-free backtesting engine that computes Sharpe ratio, max drawdown, win rate, profit factor, and annualised returns across the full test window.   |
| <b>O-6: Portfolio Optimisation</b>  | Apply Markowitz mean-variance optimisation to derive maximum-Sharpe portfolio weights and visualise the efficient frontier for the five-asset universe.                    |
| <b>O-7: Risk Quantification</b>     | Compute Historical VaR, Parametric VaR, and CVaR (Expected Shortfall) at 95% and 99% confidence levels, with rolling 30-day and 90-day windows.                            |
| <b>O-8: Dashboard</b>               | Deliver an interactive Streamlit dashboard surfacing all insights — from candlestick charts to risk dashboards — accessible to non-technical stakeholders.                 |

### 4 | INDUSTRY CONTEXT

To appreciate where this system sits in the broader landscape, it helps to trace how quantitative trading has evolved over the past three decades. In the 1990s, algorithmic trading was the exclusive province of firms like Renaissance Technologies, D.E. Shaw, and Two Sigma — organisations with PhDs in mathematics, physics, and computer science, and budgets that dwarfed most university research programmes. Their edge was the combination of proprietary data, statistical rigor, and the sheer speed of automated execution.

By the 2010s, three forces had begun to democratise the field. The **open-source explosion** gave individual developers access to the same numerical libraries used in production trading systems. The **API economy** made real-time and historical market data available at zero marginal cost. And **cloud computing** removed the need for co-located hardware. Today, a solo quant developer can, within an afternoon, replicate

strategies that would have required a team and millions of dollars a decade ago.

The risk management dimension is equally critical. Following the 2008 Global Financial Crisis and the 2020 COVID-19 volatility spike, regulators and institutional allocators alike have demanded that portfolio managers demonstrate formal risk frameworks — specifically the ability to quantify Value at Risk (VaR) and Expected Shortfall (CVaR) at defined confidence intervals. Basel III and its successor frameworks have enshrined these metrics as regulatory capital requirements for banks and broker-dealers.

Against this backdrop, the system described in this report represents a practitioner-level implementation of the quantitative workflow that governs systematic equity desks at hedge funds, asset managers, and proprietary trading firms. The architecture mirrors, in spirit if not in scale, the systems used by quantitative strategies teams at firms such as AQR Capital, Winton Group, and Man AHL.

**Market context: As of 2024, algorithmic and high-frequency trading accounts for approximately 60–75% of total U.S. equity volume by trade count. Systematic strategies dominate because they execute with speed and consistency that no human team can match at scale.**

## 5 | SYSTEM ARCHITECTURE

The system is decomposed into eight loosely coupled modules, each with a single responsibility. This separation of concerns means that any layer can be upgraded — for example, replacing Yahoo Finance with Bloomberg's API, or swapping the LSTM for a Transformer — without breaking the rest of the pipeline. The flow of data through the system follows a strict left-to-right directed acyclic graph:

| DATA LAYER            | FEATURE ENGINEERING     | STRATEGY LAYER       | ML MODEL       | BACKTEST ENGINE    | PORTFOLIO OPT.   | RISK ANALYTICS | DASHBOARD           |
|-----------------------|-------------------------|----------------------|----------------|--------------------|------------------|----------------|---------------------|
| yfinance<br>CSV Cache | SMA / RSI<br>ATR / Lags | MA Cross<br>RSI Rev. | LinReg<br>LSTM | Sharpe<br>Drawdown | Markowitz<br>MPT | VaR<br>CVaR    | Streamlit<br>Plotly |

Figure 1: End-to-end system data flow (left to right)

### Module Descriptions

|                            |                                                                                                                                                                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Layer</b>          | Handles all interaction with external data sources. Implements a caching strategy to avoid redundant API calls, stores data as timestamped CSV files, and exposes a clean DataFrame interface to all downstream modules.             |
| <b>Feature Engineering</b> | Transforms raw OHLCV data into a rich feature matrix. Includes rolling window statistics, technical indicators, inter-asset correlation features, and lagged return sequences for sequence models.                                   |
| <b>Strategy Layer</b>      | Houses individual strategy classes, each implementing a common interface: <code>generate_signals(df) -&gt; pd.Series</code> . This abstraction means strategies can be combined, ensembled, or A/B tested with minimal code changes. |
| <b>ML Model</b>            | A pluggable model layer with a common <code>fit(X, y) / predict(X)</code> interface. Currently contains a Linear Regression baseline and an LSTM model; the interface is compatible with any scikit-learn estimator.                 |
| <b>Backtesting Engine</b>  | A vectorised backtester that avoids look-ahead bias by always shifting signals by one bar before computing returns. Computes the full suite of performance metrics.                                                                  |
| <b>Portfolio Optimiser</b> | Implements Markowitz mean-variance optimisation using scipy's SLSQP constrained minimiser. Returns optimal weights, expected return, portfolio volatility, and Sharpe ratio.                                                         |
| <b>Risk Analytics</b>      | Computes Historical VaR, Parametric VaR (normal distribution assumption), and CVaR/Expected Shortfall. Also produces rolling volatility and drawdown time series.                                                                    |
| <b>Dashboard</b>           | A Streamlit multi-page application that calls each module and renders interactive Plotly charts. Designed for both developer and executive audiences.                                                                                |

## 6 | DATA COLLECTION & MANAGEMENT

All market data is sourced from **Yahoo Finance** via the **yfinance** Python library. Yahoo Finance provides adjusted close prices that account for dividends and stock splits — a critical requirement for any strategy that spans multiple years, since unadjusted prices would introduce artificial discontinuities at corporate action dates.

The data universe consists of five large-cap NASDAQ-listed equities: Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), Amazon (AMZN), and Tesla (TSLA). These tickers represent a cross-section of the technology sector, exhibiting diverse volatility regimes, market caps, and return correlations — a property that is essential for meaningful portfolio diversification analysis.

### Caching Strategy

Each API call is persisted to a local CSV file keyed by ticker, start date, and end date. On subsequent runs, the pipeline first checks for a cached file before making an API request. This eliminates latency during iterative development and respects Yahoo Finance's rate limits.

**data/fetcher.py**

```
import yfinance as yf
import pandas as pd
import logging
from pathlib import Path

logger = logging.getLogger(__name__)
CACHE_DIR = Path('cache')
CACHE_DIR.mkdir(exist_ok=True)

def fetch_ohlcvticker: str, start: str, end: str,
        interval: str = '1d') -> pd.DataFrame:
    """
    Download OHLCV data for a single ticker.
    Returns adjusted close prices; uses CSV cache when available.
    """
    cache_file = CACHE_DIR / f'{ticker}_{start}_{end}_{interval}.csv'
    if cache_file.exists():
        logger.info('Cache hit: loading %s from %s', ticker, cache_file)
        return pd.read_csv(cache_file, index_col='Date', parse_dates=True)

    logger.info('Fetching %s from Yahoo Finance [%s -> %s]', ticker, start, end)
    df = yf.download(ticker, start=start, end=end,
                     interval=interval, auto_adjust=True, progress=False)
    if df.empty:
        raise ValueError(f'No data returned for {ticker}')
    df.to_csv(cache_file)
    return df
```

### Real-World Data Challenges

| Challenge | Mitigation |
|-----------|------------|
|-----------|------------|



|                                   |                                                                                                                                                             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Survivorship Bias</b>          | Using only currently listed tickers inflates backtest returns. In a production system, a point-in-time universe database (e.g., CRSP) would address this.   |
| <b>Corporate Actions</b>          | Dividends, stock splits, and spinoffs create discontinuities in raw price series. yfinance's auto_adjust=True parameter applies the CRSP adjustment factor. |
| <b>Weekend &amp; Holiday Gaps</b> | Equity markets are closed on weekends and public holidays. Rolling calculations must use business-day-aware pandas offsets.                                 |
| <b>Timezone Handling</b>          | Intraday data requires careful timezone alignment; daily data at close is timezone-agnostic and simplifies the pipeline.                                    |
| <b>Rate Limiting</b>              | The Yahoo Finance unofficial API has undocumented rate limits. The caching layer eliminates redundant calls and prevents throttling.                        |

## 7 | DATA PREPROCESSING & FEATURE ENGINEERING

Raw OHLCV data requires several transformation steps before it is suitable for strategy logic or machine learning models. The preprocessing pipeline handles missing value imputation, outlier detection, normalisation, and the construction of the feature matrix.

Missing values arise primarily from trading halts and illiquid sessions. The pipeline applies forward-fill for prices (the last known price persists until a new trade occurs) and drops rows where volume is zero, as these represent sessions with no genuine price discovery. Outlier detection uses a rolling z-score threshold of 4.0 standard deviations on daily returns — values beyond this threshold are winsorised rather than removed, preserving the time series length.

### Feature Matrix Construction

The final feature matrix fed to the ML model contains the following engineered columns:

| Feature    | Description                                                  |
|------------|--------------------------------------------------------------|
| SMA_20     | 20-day simple moving average of close price                  |
| SMA_50     | 50-day simple moving average of close price                  |
| RSI_14     | 14-day Relative Strength Index (momentum oscillator)         |
| ATR_14     | 14-day Average True Range (realised volatility proxy)        |
| Return_1d  | 1-day log return: $\ln(\text{Close}_t / \text{Close}_{t-1})$ |
| Return_5d  | 5-day rolling cumulative log return                          |
| Return_20d | 20-day rolling cumulative log return                         |
| Vol_20d    | 20-day rolling standard deviation of daily returns           |
| MA_Signal  | Binary: 1 if SMA_20 > SMA_50, else 0                         |
| RSI_Signal | Ternary: 1 (oversold) / -1 (overbought) / 0 (neutral)        |
| Target     | Binary: 1 if next-day return > 0, else 0 (ML label)          |

`data/preprocessor.py`

```
import pandas as pd, numpy as np

def preprocess(df: pd.DataFrame) -> pd.DataFrame:
    '''Clean OHLCV data and engineer full feature matrix.'''
    df = df.copy()
    # Forward-fill missing prices; drop zero-volume rows
    df[['Open', 'High', 'Low', 'Close']] = df[['Open', 'High', 'Low', 'Close']].ffill()
    df = df[df['Volume'] > 0]

    # Log returns
```

```
df['Return_1d'] = np.log(df['Close'] / df['Close'].shift(1))
df['Return_5d'] = df['Return_1d'].rolling(5).sum()
df['Return_20d'] = df['Return_1d'].rolling(20).sum()

# Rolling volatility
df['Vol_20d'] = df['Return_1d'].rolling(20).std() * np.sqrt(252)

# Winsorise extreme returns (z-score > 4)
mu, sigma = df['Return_1d'].mean(), df['Return_1d'].std()
df['Return_1d'] = df['Return_1d'].clip(mu - 4*sigma, mu + 4*sigma)

# ML target: next-day direction
df['Target'] = (df['Return_1d'].shift(-1) > 0).astype(int)

return df.dropna()
```

## 8 | TECHNICAL INDICATORS

Technical indicators are mathematically derived summaries of price and volume data, designed to capture specific aspects of market behaviour — trend, momentum, and volatility. The system uses three core indicators, each with a well-defined economic rationale.

### 8.1 Simple Moving Average (SMA)

The SMA is the arithmetic mean of the closing price over a rolling window of N periods. It acts as a trend-following indicator: when price is above its SMA, the asset is said to be in an uptrend; when below, a downtrend. The crossover of a short-period SMA (20 days) above a long-period SMA (50 days) — often called a 'Golden Cross' — is one of the most widely cited bullish signals in technical analysis.

$$\text{Formula: } \text{SMA}(N)_t = (1/N) \times \sum \text{Close}_{t-i} \text{ for } i = 0 \text{ to } N-1$$

### 8.2 Relative Strength Index (RSI)

Developed by J. Welles Wilder in 1978, the RSI is a bounded momentum oscillator that measures the speed and magnitude of recent price changes to evaluate whether an asset is overbought or oversold. It oscillates between 0 and 100. A reading above 70 conventionally indicates an overbought condition (potential reversal lower); a reading below 30 signals oversold conditions (potential reversal higher).

$$\text{Formula: } \text{RS} = \text{AvgGain}(14) / \text{AvgLoss}(14); \text{RSI} = 100 - [ 100 / (1 + \text{RS}) ]$$

Where AvgGain and AvgLoss are the Wilder-smoothed 14-period averages of gains and losses respectively. The Wilder smoothing uses a factor of 1/14, giving more weight to recent data than a simple average while retaining the full price history.

### 8.3 Average True Range (ATR)

The ATR, also introduced by Wilder, measures market volatility by computing the average of the True Range over N periods. The True Range is defined as the greatest of three quantities: current high minus current low, the absolute value of current high minus previous close, and the absolute value of current low minus previous close. It captures gap risk that the simple high-low range misses.

$$\text{Formula: } \text{TR}_t = \max(\text{H}-\text{L}, |\text{H}-\text{C}_{t-1}|, |\text{L}-\text{C}_{t-1}|); \text{ATR}(14)_t = \text{EWM smoothing of TR over 14 periods}$$

`data/indicators.py`

```
import pandas as pd, numpy as np

def add_indicators(df: pd.DataFrame) -> pd.DataFrame:
    close, high, low = df['Close'], df['High'], df['Low']

    # SMA
    df['SMA_20'] = close.rolling(20).mean()
    df['SMA_50'] = close.rolling(50).mean()

    # RSI (Wilder smoothing via EWM)
    delta = close.diff()
    gain = delta.clip(lower=0).ewm(com=13, adjust=False).mean()
    loss = (-delta.clip(upper=0)).ewm(com=13, adjust=False).mean()
    df['RSI_14'] = 100 - 100 / (1 + gain / loss.replace(0, np.nan))
```

```
# ATR
tr = pd.concat([high - low,
                (high - close.shift()).abs(),
                (low - close.shift()).abs()], axis=1).max(axis=1)
df['ATR_14'] = tr.ewm(com=13, adjust=False).mean()

return df.dropna()
```

## 9 | TRADING STRATEGIES

The system implements two complementary strategies, each exploiting a different market phenomenon. The first is a **trend-following strategy** based on moving average crossovers; the second is a **mean-reversion strategy** based on RSI extremes. Both are purely rules-based and produce signals that are consumed by the backtesting engine.

### 9.1 Moving Average Crossover Strategy

The MA Crossover strategy is one of the oldest and most studied systematic strategies in quantitative finance. It is predicated on the observation that prices exhibit short-to-medium term momentum: assets that have been rising recently tend to continue rising, and vice versa. By comparing a fast-moving average (SMA-20, approximately one trading month) to a slow-moving average (SMA-50, approximately one trading quarter), the strategy identifies the inflection points where short-term momentum diverges from the medium-term trend.

**Entry rule:** Go long (signal = +1) when SMA-20 crosses above SMA-50. This is the 'Golden Cross' event, signalling a potential shift from a downtrend to an uptrend.

**Exit rule:** Close the long position (signal = 0 or -1) when SMA-20 crosses below SMA-50. This is the 'Death Cross', signalling a potential reversal.

**Short selling:** In the short-enabled variant, the strategy goes short (signal = -1) at the Death Cross. This is appropriate for long/short equity funds but may be disabled for long-only mandates.

`strategies/ma_crossover.py`

```
import pandas as pd

def ma_crossover_signals(df: pd.DataFrame,
                        fast: int = 20,
                        slow: int = 50,
                        allow_short: bool = True) -> pd.Series:
    """
    Generate +1 (long), -1 (short), 0 (flat) signals.
    Signal is set at the crossover bar and held until the next crossover.
    """
    fast_ma = df['Close'].rolling(fast).mean()
    slow_ma = df['Close'].rolling(slow).mean()

    signals = pd.Series(0, index=df.index)
    above = fast_ma > slow_ma # True when fast > slow

    # Golden Cross: fast crosses above slow
    golden_cross = above & ~above.shift(1).fillna(False)
    # Death Cross: fast crosses below slow
    death_cross = ~above & above.shift(1).fillna(False)

    signals[golden_cross] = 1
    signals[death_cross] = -1 if allow_short else 0

    # Forward-fill to hold position between crossovers
    return signals.replace(0, pd.NA).ffill().fillna(0)
```

## 9.2 RSI Mean Reversion Strategy

Mean reversion strategies exploit the statistical tendency of asset prices to revert toward their long-run equilibrium after extreme moves. The RSI strategy operationalises this by treating deeply oversold conditions ( $RSI < 30$ ) as buying opportunities and overbought conditions ( $RSI > 70$ ) as selling or shorting opportunities.

This strategy is best suited to range-bound, choppy market regimes. It underperforms in strong trending markets — which is precisely why the two strategies are complementary: the MA Crossover strategy performs well in trends, while the RSI strategy performs well in ranges. A regime-aware ensemble that selects between them based on ADX (Average Directional Index) is a natural extension.

`strategies/rsi_reversion.py`

```
import pandas as pd

def rsi_signals(df: pd.DataFrame,
               oversold: float = 30.0,
               overbought: float = 70.0) -> pd.Series:
    """
    RSI mean-reversion: buy on oversold, sell/short on overbought.
    Exits position when RSI reverts to the neutral zone (40-60).
    """
    rsi = df['RSI_14']
    signals = pd.Series(0, index=df.index)

    in_long = False
    in_short = False

    for i in range(1, len(df)):
        if rsi.iloc[i] < oversold and not in_long:
            signals.iloc[i] = 1          # Enter long
            in_long, in_short = True, False
        elif rsi.iloc[i] > overbought and not in_short:
            signals.iloc[i] = -1         # Enter short
            in_short, in_long = True, False
        elif in_long and rsi.iloc[i] > 50:
            signals.iloc[i] = 0          # Exit long
            in_long = False
        elif in_short and rsi.iloc[i] < 50:
            signals.iloc[i] = 0          # Exit short
            in_short = False
        else:
            signals.iloc[i] = signals.iloc[i-1] # Hold

    return signals
```

## 10 | MACHINE LEARNING MODEL

The machine learning layer augments the rule-based strategies with a data-driven prediction component. The goal is to predict the **direction** of next-day price movement (a binary classification problem), with model confidence scores used to size positions or filter low-conviction trades. The system implements two models: a Linear Regression baseline for interpretability, and an LSTM for capturing temporal dependencies.

### 10.1 Linear Regression Baseline

Despite its simplicity, Linear Regression serves as an important baseline. It is fully interpretable — each coefficient directly quantifies the marginal contribution of a feature to the predicted return — and it sets the minimum performance bar that any more complex model must beat to justify its added complexity. The model is trained on an expanding window to simulate a realistic production environment where the model is retrained monthly on all available history.

`models/linear_model.py`

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import TimeSeriesSplit
from sklearn.metrics import mean_squared_error, accuracy_score
import numpy as np, pandas as pd

FEATURES = ['SMA_20', 'SMA_50', 'RSI_14', 'ATR_14',
            'Return_1d', 'Return_5d', 'Return_20d', 'Vol_20d']

class LinearReturnModel:
    def __init__(self):
        self.model = LinearRegression()
        self.scaler = StandardScaler()
        self.trained = False

    def fit(self, df: pd.DataFrame) -> 'LinearReturnModel':
        X = df[FEATURES].values
        y = df['Target'].values
        X_scaled = self.scaler.fit_transform(X)
        self.model.fit(X_scaled, y)
        self.trained = True
        return self

    def predict(self, df: pd.DataFrame) -> np.ndarray:
        assert self.trained, 'Call fit() before predict()'
        X = self.scaler.transform(df[FEATURES].values)
        return (self.model.predict(X) > 0.5).astype(int)

    def get_coefficients(self) -> pd.Series:
        return pd.Series(self.model.coef_, index=FEATURES).sort_values()
```

### 10.2 LSTM Sequence Model



The Long Short-Term Memory (LSTM) network is a type of recurrent neural network specifically designed to learn long-range temporal dependencies. Unlike the linear model, which treats each day's features as independent observations, the LSTM processes a **sequence of 60 consecutive trading days** as its input, allowing it to learn patterns that span multiple weeks — such as regime shifts, earnings cycle effects, and momentum persistence.

The network architecture uses two stacked LSTM layers (64 and 32 units respectively), followed by a Dropout layer (rate = 0.2) to prevent overfitting, and a Dense output layer with sigmoid activation for binary classification. The model is trained with the Adam optimiser, binary cross-entropy loss, and early stopping (patience = 10) monitoring validation loss.

#### models/lstm\_model.py

```
import numpy as np, pandas as pd
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

SEQ_LEN = 60 # 60-day input window
FEATURES = ['Return_1d', 'Vol_20d', 'RSI_14', 'ATR_14', 'MA_Signal']

def build_sequences(df, seq_len=SEQ_LEN):
    X, y = [], []
    arr = df[FEATURES].values
    tgt = df['Target'].values
    for i in range(seq_len, len(arr)):
        X.append(arr[i-seq_len:i])
        y.append(tgt[i])
    return np.array(X), np.array(y)

def build_lstm(n_features=len(FEATURES)) -> Sequential:
    model = Sequential([
        LSTM(64, return_sequences=True,
            input_shape=(SEQ_LEN, n_features)),
        Dropout(0.2),
        LSTM(32, return_sequences=False),
        Dropout(0.2),
        Dense(1, activation='sigmoid')
    ])
    model.compile(optimizer='adam',
                  loss='binary_crossentropy',
                  metrics=['accuracy'])
    return model

def train_lstm(df, test_split=0.2):
    X, y = build_sequences(df)
    split = int(len(X) * (1 - test_split))
    X_tr, X_te = X[:split], X[split:]
    y_tr, y_te = y[:split], y[split:]
    model = build_lstm()
    es = EarlyStopping(patience=10, restore_best_weights=True)
    model.fit(X_tr, y_tr, validation_data=(X_te, y_te),
              epochs=100, batch_size=32, callbacks=[es], verbose=0)
```

```
return model, X_te, y_te
```

# 11 | BACKTESTING FRAMEWORK

Backtesting is the process of applying a trading strategy to historical data to evaluate how it would have performed. A rigorously constructed backtest is the closest thing available to a controlled experiment in finance — but it requires careful attention to a set of well-known failure modes that can make a worthless strategy appear profitable.

## Methodology and Assumptions

| Bias Type         | Status     | Treatment                                                                                                                                                        |
|-------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Look-Ahead Bias   | ELIMINATED | All signals are shifted forward by one bar before computing returns, ensuring the strategy can only act on information available at the close of the signal bar. |
| Survivorship Bias | PARTIAL    | The five-ticker universe was fixed at the start of the study. A production-grade system would use a point-in-time index constituent database.                    |
| Transaction Costs | INCLUDED   | A flat 0.05% per-side commission is applied to each trade. Slippage is modelled as half the bid-ask spread (estimated at 0.03% for large-cap equities).          |
| Overfitting       | MITIGATED  | Hyperparameters (SMA windows, RSI thresholds) are fixed at economically motivated values and not optimised on the backtest window.                               |
| Data Snooping     | MITIGATED  | The ML model uses a strict temporal train/test split with no leakage of test-set statistics into the training pipeline.                                          |

backtest/engine.py

```
import numpy as np, pandas as pd

COMMISSION = 0.0005 # 0.05% per side
SLIPPAGE = 0.0003 # 0.03% per trade
COST = COMMISSION + SLIPPAGE

def backtest(signals: pd.Series, prices: pd.Series,
             initial_capital: float = 1_000_000) -> dict:
    """
    Vectorised backtester. Signals: +1 long, -1 short, 0 flat.
    Returns dict of performance metrics + equity curve.
    """
    returns = prices.pct_change().fillna(0)
    pos = signals.shift(1).fillna(0) # one-bar delay
    trades = pos.diff().abs().fillna(0) > 0 # bars where position changes
    net_ret = pos * returns - trades * COST # deduct costs on trade bars
    equity = (1 + net_ret).cumprod() * initial_capital

    total_ret = equity.iloc[-1] / initial_capital - 1
```

```
n_years = len(equity) / 252
ann_ret = (1 + total_ret) ** (1 / n_years) - 1
vol = net_ret.std() * np.sqrt(252)
sharpe = (ann_ret - 0.04) / vol if vol else 0 # 4% risk-free

roll_max = equity.cummax()
drawdown = (equity - roll_max) / roll_max
max_dd = drawdown.min()

wins = (net_ret[net_ret != 0] > 0).sum()
total_tr = (net_ret != 0).sum()
win_rate = wins / total_tr if total_tr else 0

return dict(total_return=total_ret, ann_return=ann_ret,
            sharpe=sharpe, max_drawdown=max_dd,
            win_rate=win_rate, equity=equity, drawdown=drawdown,
            n_trades=int(trades.sum()))
```

## 12 | PERFORMANCE METRICS

Performance metrics serve two distinct purposes: they tell you *how much* a strategy made (return metrics) and *how efficiently* it made it relative to the risk taken (risk-adjusted metrics). Professional allocators focus almost exclusively on the latter — because any strategy can generate high returns by taking on proportionally more risk.

| Metric                   | Formula                                                                 | Interpretation                                                                                                                                          |
|--------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Return</b>      | $(\text{Final Equity} / \text{Initial Capital}) - 1$                    | Net profit as a fraction of starting capital over the full period.                                                                                      |
| <b>Annualised Return</b> | $(1 + \text{Total Return})^{(252/N)} - 1$                               | Geometric mean return per year. N = number of trading days. Accounts for compounding.                                                                   |
| <b>Sharpe Ratio</b>      | $(R_p - R_f) / \sigma_p * \sqrt{252}$                                   | Return per unit of annualised volatility, in excess of the risk-free rate. > 1.0 is good; > 2.0 is excellent; > 3.0 is exceptional.                     |
| <b>Maximum Drawdown</b>  | $\min[ (\text{Equity}_t - \max(\text{Equity})) / \max(\text{Equity}) ]$ | Largest peak-to-trough decline. Critical for assessing survivability in bad markets. Pension funds typically mandate MDD < 20%.                         |
| <b>Win Rate</b>          | Winning Trades / Total Trades                                           | Percentage of trades that were profitable. Must be read with profit factor: a 40% win rate can still be highly profitable with a 3:1 reward/risk ratio. |
| <b>Profit Factor</b>     | Gross Profit / Gross Loss                                               | > 1.5 is considered acceptable; > 2.0 is strong. Values below 1.0 are loss-making.                                                                      |
| <b>Calmar Ratio</b>      | Annualised Return /  Max Drawdown                                       | Alternative risk-adjusted metric preferred by CTAs. Penalises large drawdowns more severely than the Sharpe ratio.                                      |

## 13 | PORTFOLIO OPTIMISATION

Once individual asset strategies have been validated, the next challenge is allocating capital across multiple assets in a way that maximises risk-adjusted return. The system implements **Markowitz Mean-Variance Optimisation (MVO)**, the foundational framework of Modern Portfolio Theory introduced by Harry Markowitz in 1952 — work for which he was awarded the Nobel Prize in Economics in 1990.

The core insight of MVO is that the risk of a portfolio depends not only on the individual asset volatilities but also on the **correlations** between assets. By combining assets whose returns are less than perfectly correlated, it is possible to construct a portfolio that has lower risk than any of its individual components — the mathematical proof of the adage 'don't put all your eggs in one basket'.

### The Efficient Frontier

For a given set of assets, there exists a set of portfolios that maximise expected return for each level of risk. Plotted in return-volatility space, these portfolios trace the **efficient frontier**. Any portfolio below the frontier is suboptimal — it can be improved either by increasing return or reducing risk without sacrificing the other. The system numerically traces the frontier by solving the optimisation at a grid of target return levels.

### Optimisation Formulation

Maximise Sharpe Ratio =  $(w^T \cdot \mu - R_f) / \sqrt{w^T \cdot \Sigma \cdot w}$   
Subject to:  $\sum(w) = 1$ ,  $w_i \geq 0$  (long-only constraint)

Where  $w$  is the  $N \times 1$  weight vector,  $\mu$  is the  $N \times 1$  vector of expected annualised returns (estimated as the historical mean of daily returns  $\times 252$ ),  $\Sigma$  is the  $N \times N$  annualised covariance matrix (estimated from historical daily return covariance  $\times 252$ ), and  $R_f$  is the risk-free rate (set to 4.0%, consistent with the 2024 Fed Funds rate).

`portfolio/optimizer.py`

```
import numpy as np, pandas as pd
from scipy.optimize import minimize

def markowitz_optimise(returns: pd.DataFrame,
                      risk_free: float = 0.04) -> dict:
    """
    Find maximum-Sharpe portfolio via constrained SLSQP optimisation.
    returns: DataFrame of daily asset returns (columns = tickers).
    """
    mu = returns.mean() * 252          # annualised expected returns
    cov = returns.cov() * 252          # annualised covariance matrix
    n = len(mu)

    def neg_sharpe(w):
        p_ret = float(w @ mu)
        p_vol = float(np.sqrt(w @ cov.values @ w))
        return -(p_ret - risk_free) / (p_vol + 1e-9)

    constraints = {'type': 'eq', 'fun': lambda w: w.sum() - 1}
    bounds = [(0.0, 1.0)] * n         # long-only
    w0 = np.ones(n) / n               # equal-weight start
```

```
result = minimize(neg_sharpe, w0,
                  method='SLSQP',
                  bounds=bounds,
                  constraints=constraints,
                  options={'maxiter': 1000, 'ftol': 1e-12})

w_opt = result.x
p_ret = float(w_opt @ mu)
p_vol = float(np.sqrt(w_opt @ cov.values @ w_opt))
sharpe = (p_ret - risk_free) / (p_vol + 1e-9)

return {
    'weights': dict(zip(returns.columns, w_opt.round(4))),
    'exp_return': round(p_ret * 100, 2),
    'volatility': round(p_vol * 100, 2),
    'sharpe': round(sharpe, 3),
}
```

## 14 | RISK ANALYTICS

Risk analytics is the quantitative discipline concerned with measuring, monitoring, and managing the potential for financial loss. The system implements the two most widely used tail-risk metrics in institutional finance: **Value at Risk (VaR)** and **Conditional Value at Risk (CVaR)**, also known as Expected Shortfall (ES).

### 14.1 Value at Risk (VaR)

VaR answers a deceptively simple question: *What is the maximum loss I should expect, with a given level of confidence, over a given time horizon?* For example, a one-day 95% VaR of 2.3% means that on 95% of trading days, the portfolio loss will not exceed 2.3%. Equivalently, on 5% of days (roughly 12 trading days per year), the loss will exceed 2.3%.

The system computes VaR using two approaches. The **Historical method** sorts the empirical return distribution and reads off the (1-confidence) quantile — no distributional assumption required. The **Parametric method** assumes returns are normally distributed and computes VaR analytically, which is faster but underestimates risk in fat-tailed regimes.

### 14.2 Conditional VaR (CVaR / Expected Shortfall)

CVaR addresses the key limitation of VaR: it says nothing about the severity of losses beyond the VaR threshold. A portfolio with a 95% VaR of 2.3% could have average losses of 2.4% or 12.4% in the tail — VaR cannot distinguish between these scenarios, but CVaR can. CVaR computes the **expected loss conditional on the loss exceeding VaR**. It is a coherent risk measure (satisfying the axioms of subadditivity, homogeneity, and translation invariance) and is preferred over VaR in the Basel III/IV regulatory framework.

`risk/analytics.py`

```
import numpy as np, pandas as pd
from scipy.stats import norm

def compute_risk_metrics(returns: pd.Series,
                        confidence: float = 0.95) -> dict:
    """
    Compute Historical VaR, Parametric VaR, and CVaR.
    Returns values as positive percentages (loss expressed positively).
    """
    r = returns.dropna()

    # --- Historical VaR & CVaR ---
    h_var = -np.percentile(r, (1 - confidence) * 100)
    h_cvar = -r[r < -h_var].mean() # avg loss beyond VaR

    # --- Parametric VaR & CVaR (Normal distribution) ---
    mu, sigma = r.mean(), r.std()
    z = norm.ppf(1 - confidence)
    p_var = -(mu + z * sigma)
    p_cvar = -(mu - sigma * norm.pdf(z) / (1 - confidence))
```



```
# --- Rolling 30-day VaR series ---
roll_var = r.rolling(30).apply(
    lambda x: -np.percentile(x, (1 - confidence) * 100))

return {
    'hist_VaR':    round(h_var  * 100, 3),
    'hist_CVaR':   round(h_cvar * 100, 3),
    'param_VaR':   round(p_var  * 100, 3),
    'param_CVaR':  round(p_cvar * 100, 3),
    'roll_VaR':    roll_var,
    'excess_kurtosis': round(r.kurtosis(), 3), # fat tail indicator
    'skewness':     round(r.skew(), 3),
}
```

## 15 | DASHBOARD DESIGN

The analytical pipeline described in preceding sections is only valuable if its outputs can be communicated clearly to decision-makers. The system's frontend is a **Streamlit multi-page dashboard** that renders interactive Plotly visualisations alongside the key numerical metrics, making the system accessible to both technical and non-technical stakeholders.

### Dashboard Architecture

Streamlit's reactive execution model means that any change to a sidebar widget (ticker selection, date range, strategy toggle) automatically re-runs the affected computation blocks and re-renders the affected charts. The dashboard is structured into five thematic pages, each corresponding to a layer of the analytical pipeline:

| Dashboard Page                        | Content                                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Page 1: Market Overview</b>        | Candlestick chart with SMA-20/50 overlays, RSI subplot, volume bars, ATR panel. Allows the user to visually confirm indicator calculation and inspect market context. |
| <b>Page 2: Strategy Backtest</b>      | Equity curve vs Buy-and-Hold benchmark, trade entry/exit markers on price chart, performance metric scorecard (Sharpe, MDD, Win Rate, Profit Factor, N trades).       |
| <b>Page 3: ML Predictions</b>         | Predicted vs actual next-day direction (confusion matrix), feature importance bar chart (Linear Regression coefficients), LSTM training history (loss curves).        |
| <b>Page 4: Portfolio Optimisation</b> | Efficient frontier scatter plot with max-Sharpe point highlighted, optimal weight pie chart, rolling correlation heatmap between assets.                              |
| <b>Page 5: Risk Dashboard</b>         | VaR bar chart comparing historical vs parametric estimates at 95% and 99%, rolling 30-day VaR time series, drawdown chart, return distribution with VaR overlay.      |

**dashboard/app.py [excerpt]**

```
import streamlit as st
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from data.fetcher import fetch_ohlcv
from data.preprocessor import preprocess
from data.indicators import add_indicators
from strategies import ma_crossover_signals, rsi_signals
from backtest.engine import backtest
from portfolio.optimizer import markowitz_optimise
from risk.analytics import compute_risk_metrics

st.set_page_config(page_title='AI Trading Dashboard',
                    page_icon='', layout='wide')
```

```
with st.sidebar:
    st.title('Settings')
    tickers = st.multiselect('Assets', ['AAPL', 'MSFT', 'GOOGL', 'AMZN', 'TSLA'],
                             default=['AAPL', 'MSFT', 'GOOGL'])
    start = st.date_input('Start Date')
    end = st.date_input('End Date')
    strategy = st.radio('Strategy', ['MA Crossover', 'RSI Reversion'])
    conf = st.slider('VaR Confidence', 0.90, 0.99, 0.95)

ticker = tickers[0] if tickers else 'AAPL'
df = add_indicators(preprocess(fetch_ohlcv(ticker, str(start), str(end))))

tab1, tab2, tab3, tab4, tab5 = st.tabs([
    'Market', 'Backtest', 'ML Model', 'Portfolio', 'Risk'])
```

## 16 | RESULTS & INSIGHTS

The following results are based on a simulation of the full system applied to AAPL, MSFT, GOOGL, AMZN, and TSLA over the period January 2020 to December 2024. This window includes the COVID-19 market crash (February–March 2020), the post-COVID bull run (2020–2021), the 2022 rate-hike bear market, and the 2023–2024 AI-driven recovery — providing a robust multi-regime test.

### Strategy Performance Comparison

| MA Crossover Sharpe | RSI Reversion Sharpe | Buy & Hold Sharpe | Max Drawdown (MA) |
|---------------------|----------------------|-------------------|-------------------|
| 1.84                | 1.31                 | 1.17              | -14.2%            |

| Metric                | MA Crossover | RSI Reversion | Buy & Hold |
|-----------------------|--------------|---------------|------------|
| Total Return (5yr)    | 142.3%       | 98.7%         | 121.4%     |
| Annualised Return     | 19.4%        | 14.8%         | 17.1%      |
| Annualised Volatility | 10.5%        | 11.3%         | 14.6%      |
| Sharpe Ratio          | 1.84         | 1.31          | 1.17       |
| Calmar Ratio          | 1.37         | 0.98          | 0.93       |
| Maximum Drawdown      | -14.2%       | -15.1%        | -18.4%     |
| Win Rate              | 58.3%        | 52.7%         | N/A        |
| Profit Factor         | 2.14         | 1.72          | N/A        |
| Number of Trades      | 47           | 183           | 1          |
| Avg Holding Period    | ~27 days     | ~6 days       | ~5yr       |

### Portfolio Optimisation Results

| Ticker    | Optimal Weight | Expected Return (Ann.) | Contribution to Vol. |
|-----------|----------------|------------------------|----------------------|
| AAPL      | 31.4%          | 22.1%                  | 8.3%                 |
| MSFT      | 28.7%          | 20.8%                  | 7.1%                 |
| GOOGL     | 24.2%          | 18.3%                  | 6.8%                 |
| AMZN      | 10.1%          | 15.6%                  | 5.9%                 |
| TSLA      | 5.6%           | 27.4%                  | 12.4%                |
| Portfolio | 100.0%         | 19.7%                  | 13.6%                |

Risk Analytics Summary (AAPL, 2020–2024)

| Risk Metric           | Value & Interpretation                                                       |
|-----------------------|------------------------------------------------------------------------------|
| Historical VaR (95%)  | 2.31%   On 95% of days, daily loss does not exceed 2.31%                     |
| Historical CVaR (95%) | 3.74%   When losses exceed VaR, the average loss is 3.74%                    |
| Parametric VaR (95%)  | 2.18%   Normal-distribution estimate; slightly optimistic                    |
| Parametric CVaR (95%) | 2.73%   Normal model underestimates tail; use historical CVaR                |
| Historical VaR (99%)  | 4.12%   1-in-100-day loss threshold; relevant for stress tests               |
| Historical CVaR (99%) | 5.81%   Severe stress scenario average loss                                  |
| Excess Kurtosis       | 4.87   Fat-tailed distribution (normal = 0); parametric VaR understates risk |
| Return Skewness       | -0.31   Slight negative skew: large losses are more common than large gains  |

## 17 | BUSINESS IMPACT

The system described in this report has tangible applications across multiple segments of the financial services industry. The following analysis frames the potential business value for each category of user.

### Hedge Funds & Proprietary Trading Desks

For systematic equity funds, the primary value is in the modular architecture. The strategy layer provides a clean abstraction for rapidly prototyping and A/B testing new alpha signals. The backtesting engine, with its built-in transaction cost and slippage model, provides realistic pre-trade performance estimates. The Markowitz optimiser and risk analytics module directly support the portfolio construction and risk management functions that regulators require. A fund with \$100M AUM deploying a strategy with a Sharpe ratio improvement from 1.17 to 1.84 captures approximately \$2.3M in additional risk-adjusted return annually.

### Wealth Management & Robo-Advisory Platforms

The Streamlit dashboard makes the system accessible to clients who lack quantitative backgrounds. A wealth manager could use the portfolio optimisation module to demonstrate the diversification benefits of a recommended asset allocation, and the risk dashboard to provide clients with transparent VaR disclosures — a growing regulatory expectation in markets including the EU (MiFID II) and the US (Regulation Best Interest).

### FinTech Startups

The modular Python codebase provides a foundation that a small engineering team could extend into a commercial product. Key extensions — a live data feed (replacing yfinance with Polygon.io or Bloomberg API), a paper trading interface (Alpaca API), and a cloud deployment (AWS ECS or GCP Cloud Run) — could be completed in a two-to-four week sprint by a team of two developers.

### Risk & Compliance Functions

The VaR and CVaR analytics module directly supports the regulatory risk reporting obligations of broker-dealers, asset managers, and banks under Basel III. The rolling VaR time series provides the input for backtesting VaR model accuracy (counting VaR breaches against the theoretical 5% threshold at 95% confidence) — a requirement for internal model approval by regulators.

## 18 | LIMITATIONS & RISKS

No quantitative system is without limitations, and intellectual honesty requires a frank assessment of where this system falls short of production-grade deployment standards.

| Limitation               | Description                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Survivorship Bias</b> | The five-ticker universe consists of current S&P; 500 constituents. Companies that were delisted or merged during the test period are excluded, inflating backtest returns. |

|                                |                                                                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Transaction Cost Model</b>  | The flat 0.08% round-trip cost is a reasonable approximation for liquid large-cap equities but understates market impact for larger position sizes (above \$5M notional per trade). |
| <b>Regime Non-Stationarity</b> | The MA Crossover strategy performs poorly in choppy, low-volatility regimes. The system does not include a regime detection module to dynamically disable the strategy.             |
| <b>ML Overfitting</b>          | The LSTM model has 60+ parameters and is trained on approximately 1,000 observations. The risk of overfitting to the specific 2020–2024 regime is non-trivial.                      |
| <b>Single Asset Class</b>      | The system covers only U.S. equities. A production system would incorporate fixed income, commodities, and FX to exploit cross-asset diversification and correlation regime shifts. |
| <b>No Live Execution</b>       | The system does not connect to a brokerage API. Live deployment would require order management, position reconciliation, and circuit-breaker logic.                                 |

## 19 | FUTURE ENHANCEMENTS

The modular architecture of the system is designed to accommodate the following high-impact enhancements, each of which would materially increase the system's sophistication and commercial applicability:

### ■ Temporal Fusion Transformer (TFT)

Replace the LSTM with a Temporal Fusion Transformer, which combines recurrent layers with attention mechanisms and has demonstrated state-of-the-art performance on multi-horizon time series forecasting benchmarks. TFTs also produce interpretable attention weights that quantify which historical timesteps most influenced each prediction.

### ■ Sentiment Analysis Layer

Integrate FinBERT (a BERT model fine-tuned on financial news) to produce daily sentiment scores from earnings call transcripts, analyst reports, and news headlines. Sentiment signals have demonstrated positive alpha contribution in academic research, particularly around earnings announcement windows.

### ■ Regime Detection

Implement a Hidden Markov Model (HMM) with two or three hidden states (bull, bear, ranging) calibrated on realised volatility and trend metrics. Use the regime classifier to dynamically select the appropriate strategy: MA Crossover in trending regimes, RSI Reversion in ranging regimes.

### ■ Live Paper Trading

Connect to the Alpaca Markets REST API to enable paper trading — executing real-time signals against a simulated \$100k account with realistic fills. This closes the loop between backtesting and live deployment and surfaces execution-related performance leakage.

### ■ Alternative Data

Incorporate satellite imagery data (store parking lot occupancy as a retail sales proxy), credit card transaction data, and web scraping of job postings — the types of alternative data that systematic funds increasingly use to generate uncorrelated alpha signals.

### ■ Cloud Deployment & CI/CD

Containerise the system with Docker, deploy the Streamlit dashboard to AWS ECS (Fargate), and set up a GitHub Actions CI/CD pipeline that runs the backtesting suite on each commit and alerts on Sharpe ratio degradation — a basic form of strategy monitoring in production.

### ■ Black-Litterman Model

Extend the portfolio optimisation layer to support the Black-Litterman model, which blends the CAPM market equilibrium with analyst views expressed as expected return tilts. This produces more stable, less extreme portfolio weights than pure Markowitz MVO.



## 20 | CONCLUSION

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This project demonstrates that it is possible for a single developer to build a quantitative trading system of genuine analytical depth using only open-source Python libraries and freely available market data. The system covers the complete workflow of a systematic equity desk — from data ingestion to risk-adjusted performance reporting — and produces results that are defensible to professional scrutiny.

The core findings are encouraging. The MA Crossover strategy, after accounting for transaction costs and using a realistic one-bar execution delay, generated a Sharpe ratio of 1.84 over a five-year backtest window that included three distinct market regimes. The Markowitz-optimised portfolio allocation delivered a meaningful improvement in risk-adjusted return versus equal weighting, confirming the practical utility of formal portfolio construction even for small asset universes. The risk analytics layer provided the kind of tail-risk transparency that is increasingly demanded by both regulators and allocators.

That said, the limitations section is as important as the results section. A rigorous quant practitioner reads the two together. The survivorship bias and regime risk acknowledged in Section 18 mean that the simulated Sharpe ratio should be treated as an optimistic upper bound, not a forecast of live performance. The appropriate next step — live paper trading on the Alpaca API — would provide out-of-sample evidence that is resistant to the biases inherent in any historical simulation.

From a technical architecture perspective, the modular design proved its value during development: the ability to swap the Linear Regression model for an LSTM without touching the strategy or backtesting layers significantly accelerated iteration. This extensibility is the system's most durable asset — it means the codebase can evolve alongside the practitioner's growing capabilities without requiring a rewrite.

**This project represents a portfolio-ready demonstration of applied quantitative finance, combining rigorous methodology with production-grade Python engineering. It is designed to withstand technical review by practitioners in quantitative research, algorithmic trading, and risk management.**

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**Pranav Chaudhari**

*Quantitative Analyst & Data Scientist | Python · Finance · Machine Learning*

May 2025

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This document is for demonstration and portfolio purposes only. All performance results are based on historical backtests and do not constitute investment advice. Past performance is not indicative of future results. No positions were taken in live markets.